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# Biological pre-treatment: Enhancing biogas production using the highly cellulolytic fungus *Trichoderma viride*

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ABSTRACT

With regard to renewable sources of energy, bioconversion of lignocellulosic biomass has long been recognized as a desirable endeavor. However, the highly heterogeneous structure of lignocellulose restricts the exploitation of its promising potential in biogas plants. Hence, effective pre-treatment methods are decisive prerequisites to overcome these challenges in order to improve the utilization ratio of (ligno) cellulosic substrates during fermentation. In the present study, the application of *Trichoderma viride* in an aerobic upstream process prior to anaerobic digestion led up to a threefold increase in the yield of methane and total gas in a lab-scale investigation. Due to its highly efficient cellulolytic activities, *T. viride* seemed to be responsible for an improved nutrient availability that positively influenced the anaerobic microbiocenosis. Aerobic pre-treatment of organic matter with *T. viride* is therefore a promising solution to achieve higher methane yields and degradation performances without any additional energy demand, nor undesired by-product inhibition.

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## 1. Introduction

Worldwide recognition starts to prevail that the current energy policy demands urgent replacement by more eco-conscious and sustainable procedures. Renewable energies contribute towards environmental compatibility, supply security, and prevention of resource-related conflicts and have therefore progressively gained global acclaim. Among potential renewable sources of energy bioconversion of lignocellulosic biomass has long been recognized as a desirable endeavor (Ziemiński et al., 2012; Bayer et al., 2007). Lignocellulosic biomass refers to a highly abundant, ubiquitous and inexhaustible source of organic matter mainly composed of cellulose (40–50%), hemicellulose (20–40%), and lignin (5–30%) with a worldwide production of 10–50 billion tons per annum, accounting for about half of the global biomass yield (Pérez et al., 2002, McKendry, 2002; Zhao et al., 2012). However, successfully overcoming the inherent recalcitrance of lignocellulose in order to enhance the substrate conversion is of utmost importance to obtain high yields and high productivity indispensable to commercial success (Wyman et al., 2005). Biomass recalcitrance is attributable to the heterogeneous structure and physicochemical complexity of lignocellulose comprehensively discussed by Mosier et al. (2005), Zheng et al. (2009), and Hendriks and Zeeman (2009). Improvements therefore rely on efficient pre-treatment methods that alter and/or remove structural impediments to hydrolysis, thus facilitating the accessibility and digestibility of (ligno) cellulosic substrates during fermentation (Mosier et al., 2005). Current pre-treatment technologies comprise physical, chemical, physicochemical, and biological methods (Sun and Cheng, 2002). Regarding chemical and physical pre-treatment methods, inherent drawbacks such as high costs and energy requirements, waste stream generation, and the formation of toxic by-products (e.g., weak acids, furan derivatives, and phenolic compounds) inhibitory to the subsequent hydrolysis and fermentation process have set barriers to industrial application (Sun and Cheng, 2002). Biological pre-treatment, however, is increasingly being advocated as an environmentally friendly process with low energy input, low disposal costs, mild operating conditions, and minimal by-product formation (Sun and Cheng, 2002). Biological pre-treatment is based on enzyme initiated cascade reactions employing (ligno) cellulolytic microorganisms, including fungi and bacteria (Dashtban et al., 2009; Chaturvedi and Verma, 2013). Among them, *Trichoderma viride* (Ascomycota, Hypocreales, Hypocreaceae) (Chaverri and Samuels, 2003) is known as a ubiquitous colonizer of cellulosic materials and prolific producer of a complex array of cellulose degrading enzymes comprising endoglucanases, cellobiohydrolases (exoglucanases), and β-glucosidases that work in a synergistic manner (Schuster and Schmoll, 2010; Do Vale et al., 2014). Despite prominent cellulolytic

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activities, *T. viride* exhibits an ensemble of carbohydrate-active enzymes in its secretome such as hemicellulases and pectinases to elicit effective hydrolysis. Due to its high pH and temperature tolerance, easy cultivability, and high level of genetic diversity (Harman et al., 2004) *T. viride* and its specific multi-enzyme system is of particular interest for lignocellulose pre-treatment applications.

## 2. Material and methods

### 2.1. Cultivation of T. viride

The organism used for the preliminary experiments was isolated during an investigation developing a compost starter kit (Illmer et al., 2007) and identified as *T. viride* by DSMZ (Deutsche Sammlung für Mikroorganimsen und Zellkulturen). *T. viride* was cultivated on malt extract agar plates (malt extract 30 g, peptone 5 g, agar 20 g, 1000 ml distilled water, pH 6.5) at 25 °C in the present study. Cultures were preserved at 4 °C for short-term maintenance. For the preparation of liquid cultures a sterile malt extract solution (malt extract 30 g, peptone 5 g, 1000 ml distilled water, supplemented with 0.1% antifoam agent, pH 6.5) was used, inoculated with the biomass extracted from three days old cultures on malt extract agar plates, and incubated at 25 °C at 150 rpm for two days. Sporogenesis was carried out on a solid matrix according to the procedure described in Illmer et al. (2007). During spore production in lab scale (workload for one person 0.5–1 day) a concentration of $1 \times 10^7$ spores $g^{-1}$ can be achieved within 14 days and used to pre-treat 0.8–1.0 t of fresh bio-waste. The spores obtained in dry powder were suspended in sterile Ringer's solution supplemented with Tween 80 and used as *T. viride* inoculum. The viable spore concentration in aqueous solution was assessed by cultivation on malt extract agar plates and cfu counting.

### 2.2. Substrate (standard bio-waste)

A mixture of organic waste representative of the average composition of municipal solid waste was used as initial substrate. The composition and characteristics of the substrate used are outlined in Tables 1 and 2. The mixed bio-waste was chopped (target particle size $\leqslant$ 2 cm), portioned, and stored frozen. Dry matter of the bio-waste was ascertained by drying at 105 °C until constant weight was obtained. Loss on ignition analysis at 550 °C was used to determine the organic matter content.

### 2.3. Experimental design

Based on previous findings (Wagner et al., 2013b) the soft rot ascomycetal fungus *T. viride* was used in the present study to pre-treat the biomass samples in an aerobic upstream process prior to anaerobic digestion. The procedure adopted aimed at improving the substrate degradability and catabolism of organic carbon compounds contained to achieve an increase in substrate specific methane and total gas yield and thus a higher efficiency and profitability during the anaerobic bio-waste treatment. The experimental setup was therefore divided into two parts: (i) aerobic pre-treatment and (ii) subsequent anaerobic digestion. With the purpose of evaluating the impact of pre-treatment on the subsequent anaerobic digestion, process-relevant parameters were studied under varied conditions (Section 2.3.1), optimized for maximum methane and gas yield, and maintained at their optimized level for further investigations.

**Table 1**
Composition of the initial substrate (standard bio-waste).

| Components | Including | kg 100 $kg^{-1}$ standard waste (fresh weight) |
|---|---|---|
| Kitchen waste | Various vegetables (30%), fruits (20%), boiled noodles (20%), salad (10%), dried bread (10%), chopped paper (9%), coffee grounds (1%) | 44 |
| Garden waste | Plant residues, tree cuttings, etc. | 20 |
| Grass clipping | | 16 |
| Foliage | | 12 |
| Wood pellets | | 8 |

**Table 2**
(Bio) chemical parameters of the standard bio-waste.

| Type of analysis | Means | ±SD |
|---|---|---|
| pH | 5.5 | 0.17 |
| Dry weight (DW) | 50.1 g 100 $g^{-1}$ | 1.36 |
| Organic matter content (OM) | 0.89 g $g^{-1}$ DW | 0.034 |
| Total carbon | 33.9 g 100 $g^{-1}$ DW | 3.30 |
| Total nitrogen | 1.53 g 100 $g^{-1}$ DW | 0.241 |
| C/N ratio | 22.2 | |

#### 2.3.1. Aerobic pre-treatment

Aerobic pre-treatment was carried out by inoculating 250 ml Schott flasks containing 25.0 g of the biomass samples (Table 1) with suspended spores of *T. viride*. An equivalent volume of inactivated spores was added to the control approaches. During aerobic pre-treatment single parameter variation studies including inoculation density, initial water content, pre-incubation period, and composition of the substrate were performed.

In order to evaluate the impact of the initial inoculation density during the aerobic pre-treatment process (Section 3.1), the standard bio-waste was supplemented with different amounts of spores ranging from $2 \times 10^6$ to $4 \times 10^8$ active spores 25 $g^{-1}$ substrate, while inactivated spores were added to the controls. The optimum moisture during aerobic pre-treatment was investigated by altering the moisture content of the biomass samples using different initial water levels (Section 3.2). With the purpose of investigating the impact of the aerobic pre-treatment duration (Section 3.3), the standard bio-waste was supplemented with $2 \times 10^8$ active and inactive spores, respectively, and adjusting the moisture content to 70% before undergoing a three-day, ten-day, and twenty-day aerobic pre-treatment. Studies focusing on the composition of the initial substrate were carried out by inoculating the biomass samples with $2 \times 10^8$ active and inactive *T. viride* spores 25 $g^{-1}$ substrate, respectively, adjusting the moisture content to 70%, and by a two- and fourfold increase in the content of wood (chopped wood $\leqslant$ 2 cm, diameter 0.5–1 cm) (Section 3.4).

During aerobic pre-treatment the Schott flasks were closed with cotton stoppers and aerobically incubated at room temperature (22 °C ± 2 °C), with samples regularly taken to assay carboxymethyl cellulase (CMCase) activity, total C and N, and the C/N ratio. CMCase activity was quantified following the instructions of Schinner et al. (1995), total carbon and nitrogen content was determined using a TruSpec® CHN analyzer (Leco, Germany).

#### 2.3.2. Anaerobic digestion

In the second step, the flasks containing the pre-treated biomass samples were airtight sealed with butyl rubber septa (Ochs, Germany) and inoculated with diluted fermenter sludge (DFS) derived from a 750.000-l biogas plant located in Roppen (Tyrol, Austria) as described in Illmer et al. (2009). Some basic parameters and running conditions can be found in Wagner et al. (2014). The DFS used as inoculum in the present investigations was prepared anaerobically as described previously (Wagner et al., 2013a). The anaerobic incubation period extended over 14 days in total at

52 °C, with samples withdrawn on day 0, 1, 2, 4, 7, 10, 12 and 14 from the bottles headspace and analyzed on $CH_4$, $CO_2$ and $H_2$ by gas chromatography on a Shimadzu GC 2010 as described in Wagner et al. (2011). An aliquot of the supernatant liquor was taken on day 0, 4, 8, 12 and 14 and submitted to HPLC to detect the volatile fatty acid (VFA) concentrations as described in Wagner et al. (2012). Overpressure was drained after temperature equilibration using a cannula after the sampling procedure was completed. At the end of the anaerobic incubation period microbial metabolic activity was determined by measuring potential dimethyl sulfoxide (DMSO) reduction as well as dehydrogenase activity according to the methods in Schinner et al. (1995). In addition, the electrical conductivity (EC) and pH-values were ascertained.

### 2.4. Statistical analysis

Statistical analysis was performed by using the Software package *Statistica 9.0* (StatSoft®) and *SigmaPlot 12.0* (Systat Software Inc.). Results are given as means ± standard deviation out of three replicates. Significant differences were ascertained by one-way or multifactorial ANOVA. A significance level of 0.05 was used to assess differences between treatments. *Fisher's Least Significant Difference Test* was used to discriminate between single variants.

## 3. Results and discussion

In the course of aerobic pre-treatment with *T. viride*, the impact of varied initial inoculation densities, moisture contents, pre-incubation periods, and compositions of the initial substrate on the subsequent anaerobic digestion was examined as specified in Section 2.3. The procedure adopted was conducted as a kind of feasibility study using a lab scale approach.

### 3.1. Impact of varied initial inoculation densities

The application of *T. viride* during aerobic pre-treatment had a significantly positive effect on the anaerobic degradation process, with a considerable increase in the yield of methane and total gas in all variants supplemented with active fungal spores compared to the controls (Fig. 1). At the beginning of the anaerobic incubation period the methane yield of the individual test approaches did not differ significantly while presenting increasingly different yields after an initial lag-phase of two days (Fig. 1A). Within 14 days of anaerobic incubation a clear distinction in cumulative methane and total gas production between the inoculated and non-inoculated samples could be observed (Fig. 1A and B). An increase in inoculation density gradually improved the $CH_4$ and gas production, with the highest outputs achieved upon using an inoculum of $2 \times 10^8$ *T. viride* spores 25 $g^{-1}$ substrate (Fig. 1A and B), resulting in an 88% $CH_4$ yield improvement compared to the controls (Fig. 1C). With a further increase in inoculation density to $4 \times 10^8$ spores 25 $g^{-1}$ substrate the methane and gas yield significantly declined (Fig. 1A and B). To enable comparison to other studies, additional data on cumulative $CH_4$ and gas production related to the organic matter content of the substrate (in connection with experimental conditions) are outlined in Table 3. In search of the major contributing factors responsible for the substantial enhancement in productivity, pH measurement did not provide any explanation since all test approaches (± inoculation with *T. viride*) exhibited pH-neutral values ranging from 7.2 to 7.4 (Table 3), thus lying within the narrow margin (pH 6.8–7.5) specified for stable methanogenesis during anaerobic digestion (Gerardi, 2003). However, the increase in the yield of methane and total gas in the samples supplemented with *T. viride* correlated with both, a considerable increase in CMCase activity at the end of the aerobic pre-treatment (Fig. 2A) as well as electrical conductivity (EC) (Fig. 2B) at the end of the anaerobic treatment. Especially the increased CMCase activity corresponding to the gradually increasing yields of soluble, reducing sugar equivalents pointed to an improved availability of nutrients, as proven by the increase in EC (Fig. 2B) and higher microbial activities, as assayed by DMSO reductase potentials (Fig. 2C), which peaked at

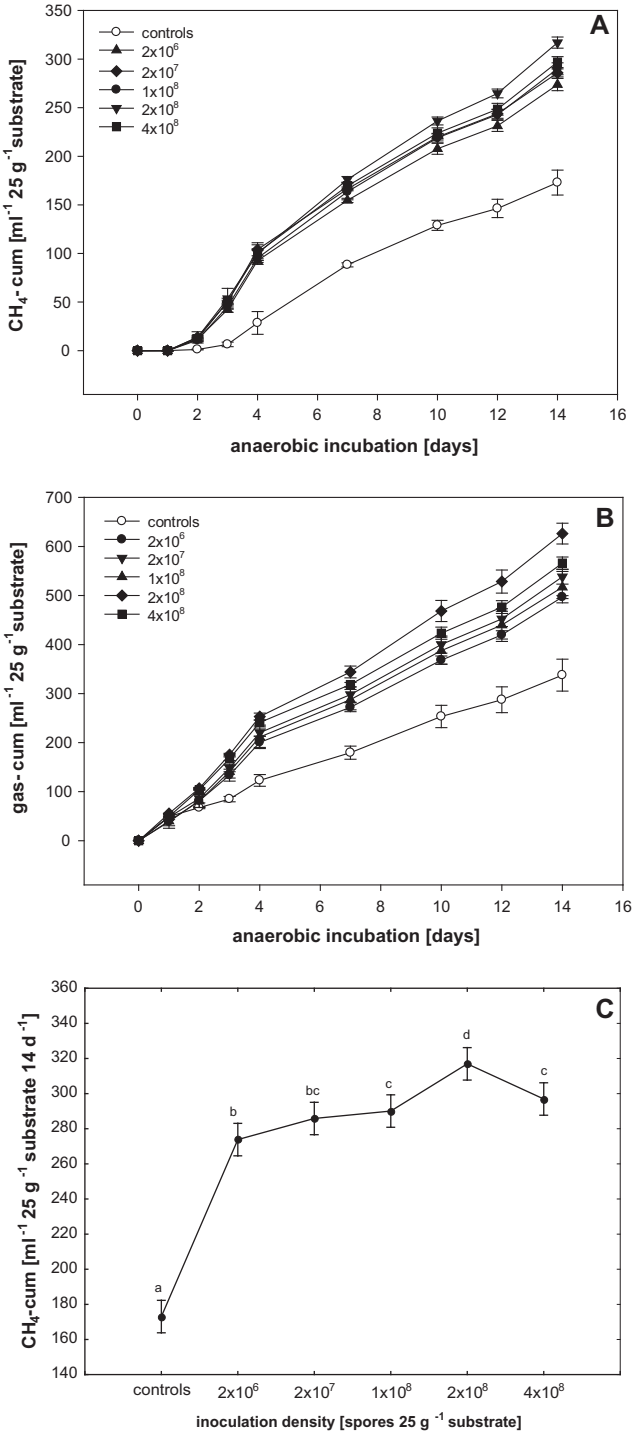



**Fig. 1.** (A) Cumulative methane and (B) total gas production as a function of time at the different inoculation ratios used. Data points are means (±SD), $n = 3$. (C) Impact of pre-treatment with *T. viride* on the cumulative methane and total gas production after 14 days of anaerobic incubation. Significant differences ($p < 0.05$) are indicated by different characters (a, b, c). Results are given as means out of three replicates. Error bars are standard deviations.

**Table 3**
Total gas and methane production related to the organic matter content as well as pH after 14 days of anaerobic digestion under varied experimental conditions. Results are given as means (±SD). Optimal conditions are highlighted.

| Experimental variation | Pre-treatment | | Cumulative $CH_4$ ($ml^{-1}$ $g^{-1}$ OM 14 $d^{-1}$) | Cumulative gas ($ml^{-1}$ $g^{-1}$ OM 14 $d^{-1}$) | pH |
|---|---|---|---|---|---|
| Inoculation density (spores 25 $g^{-1}$ substrate) | Control | Without | 194.4 (13.70) | 379.5 (36.45) | 7.2 (0.04) |
| | *T. viride* | $2 \times 10^6$ | 307.7 (8.95) | 558.3 (3.32) | 7.2 (0.02) |
| | *T. viride* | $2 \times 10^7$ | 321.2 (5.54) | 581.0 (35.94) | 7.3 (0.06) |
| | *T. viride* | $1 \times 10^8$ | 325.9 (9.33) | 604.8 (16.99) | 7.2 (0.01) |
| | *T. viride* | **$2 \times 10^8$** | **356.1 (9.10)** | **703.7 (23.80)** | **7.3 (0.06)** |
| | *T. viride* | $4 \times 10^8$ | 333.6 (8.32) | 363.0 (14.07) | 7.3 (0.06) |
| Initial moisture content (%) | Control | 50 | 284.2 | 491.4 | 7.2 |
| | Control | 70 | 314.2 | 553.7 | 7.2 |
| | Control | 90 | 352.1 | 618.3 | 7.2 |
| | *T. viride* | 50 | 355.4 (24.81) | 625.4 (36.47) | 7.2 (0.01) |
| | *T. viride* | **70** | **447.7 (6.36)** | **790.0 (16.91)** | **7.3 (0.02)** |
| | *T. viride* | 90 | 473.7 (28.26) | 810.0 (47.15) | 7.3 (0.08) |
| Aerobic pre-incubation (days) | Control | 3 | 1.3 (0.11) | 231.4 (20.59) | 6.1 (0.03) |
| | Control | 10 | 133.3 (85.02) | 367.4 (178.25) | 7.2 (0.02) |
| | Control | 20 | 214.4 (12.26) | 372.6 (15.90) | 7.2 (0.04) |
| | *T. viride* | 3 | 4.7 (0.85) | 293.6 (32.39) | 6.2 (0.07) |
| | *T. viride* | **10** | **439.5 (58.20)** | **840.9 (114.04)** | **7.3 (0.03)** |
| | *T. viride* | 20 | 315.2 (23.42) | 525.2 (5.33) | 7.3 (0.04) |
| Substrate composition (increase in wood content) | Control | Unmod[a] | 312.3 (15.89) | 688.3 (22.83) | 7.1 (0.04) |
| | Control | 2-fold | 225.2 (23.34) | 479.9 (31.60) | 7.0 (0.11) |
| | Control | 4-fold | 149.6 (20.26) | 341.4 (8.13) | 6.9 (0.02) |
| | *T. viride* | **Unmod[a]** | **722.6 (66.87)** | **1299.4 (57.55)** | **7.3 (0.03)** |
| | *T. viride* | 2-fold | 516.9 (31.28) | 989.2 (64.50) | 7.3 (0.03) |
| | *T. viride* | 4-fold | 377.4 (26.51) | 744.1 (49.35) | 7.2 (0.04) |

[a] Unmodified.

initial inoculation densities of $2 \times 10^8$ spores 25 $g^{-1}$ substrate. Maximum CMCase yield was achieved after ten days of aerobic pre-treatment and again with an inoculation density of $2 \times 10^8$ spores 25 $g^{-1}$ substrate (Fig. 2A), coinciding with the significant increase in methane yield obtained during gas quantification. These results suggested that high cellulolytic activities of *T. viride* during aerobic pre-treatment seemed to be responsible for improved substrate conversion rates inducing a sufficient nutrient supply for the methane forming microbiocenosis, and thus reflecting the positive influence of pre-treatment on gas yield within anaerobic treatment. The observed yield loss after ten days of pre-treatment and $4 \times 10^8$ spores 25 $g^{-1}$ substrate, however, might be related to overcrowding of spores causing an inhibition of germination and development, "catabolite repression" of the cellulase system or due to the depletion of macro- and micronutrients altering the fungal physiology (Nochure et al., 1993; Ghanem et al., 2000). Another possible reason is that the high abundance of metabolic active microorganisms associated with an increased total consumption of solubilized organic matter caused the net solubilization yields to cease (Fernández-Güelfo et al., 2011), resulting in a significant decrease in CMCase activity, electrical conductivity as well as decreasing metabolic activities as determined by measuring potential DMSO reduction (Fig. 2C). This in turn might have caused $CH_4$ yields to significantly cease during the anaerobic incubation in the samples supplemented with $4 \times 10^8$ spores 25 $g^{-1}$ substrate compared to those supplemented with $2 \times 10^8$ *T. viride* spores 25 $g^{-1}$ substrate. In summary, the inoculation density emerged as a decisive factor exerting a highly positive and significant impact on the overall process efficiency, with the highest outputs achieved upon using an inoculum with $2 \times 10^8$ active fungal spores 25 $g^{-1}$ substrate, and was thus considered sufficient for further investigations.

### 3.2. Impact of different initial moisture contents

In accordance with the results obtained, the inoculation with *T. viride* during aerobic pre-treatment once again exerted a significant impact on the methane production rates within anaerobic treatment, resulting in higher methane and gas yields in all variants supplemented with active fungal spores compared to those without inoculation (Fig. 3A and B). Moreover, a distinct influence of the initial water content could be observed. While the application of *T. viride* during aerobic pre-treatment only caused a marginal increase of 30% in cumulative methane generation during anaerobic treatment in the samples with a moisture content of merely 50%, the methane yields significantly increased with increasing moisture level, resulting in a 50% yield improvement in the samples with a moisture content of 70% and *T. viride* pre-treatment compared to the yield noted in the corresponding controls (Fig. 3C). Although a further increase in moisture level resulted in less of an increase (33%) compared to the controls, fungal pre-treatment of the biomass samples with an initial moisture content of 90% led to the highest methane yield after 14 days of anaerobic incubation, however, no significant difference to the samples with 70% moisture and *T. viride* pre-treatment could be ascertained (Fig. 3C). Again neutral pH- values within a narrow margin of 7.2–7.4 were observed at the end of the anaerobic treatment, concluding that the pH level had no significant influence on gas yield (Table 3). Nevertheless, increasing initial moisture contents showed a positive effect on CMCase activity related to increasing yields of reducing sugar equivalents, with maximum activities occurring in the samples with 70% and 90% moisture and *T. viride* pre-treatment (Fig. 4A). Increasing moisture levels further resulted in an increase in EC in both inoculated and non-inoculated samples with a significant increase in EC in the samples with initial water contents of 70% and 90% at the end of anaerobic incubation (Fig. 4B), which is particularly interesting since higher water levels induce a higher degree of dilution. This seemed to imply that *T. viride* initiates ion concentrations and hence nutrient availability to increase at favorable water conditions. The increase in CMCase activities along with increasing moisture might be related to the fact that species of the genus *Trichoderma* are considered as "hydrophilic fungi" ($a_w$ higher than 0.9) favoring moist habitats and substrates with high water

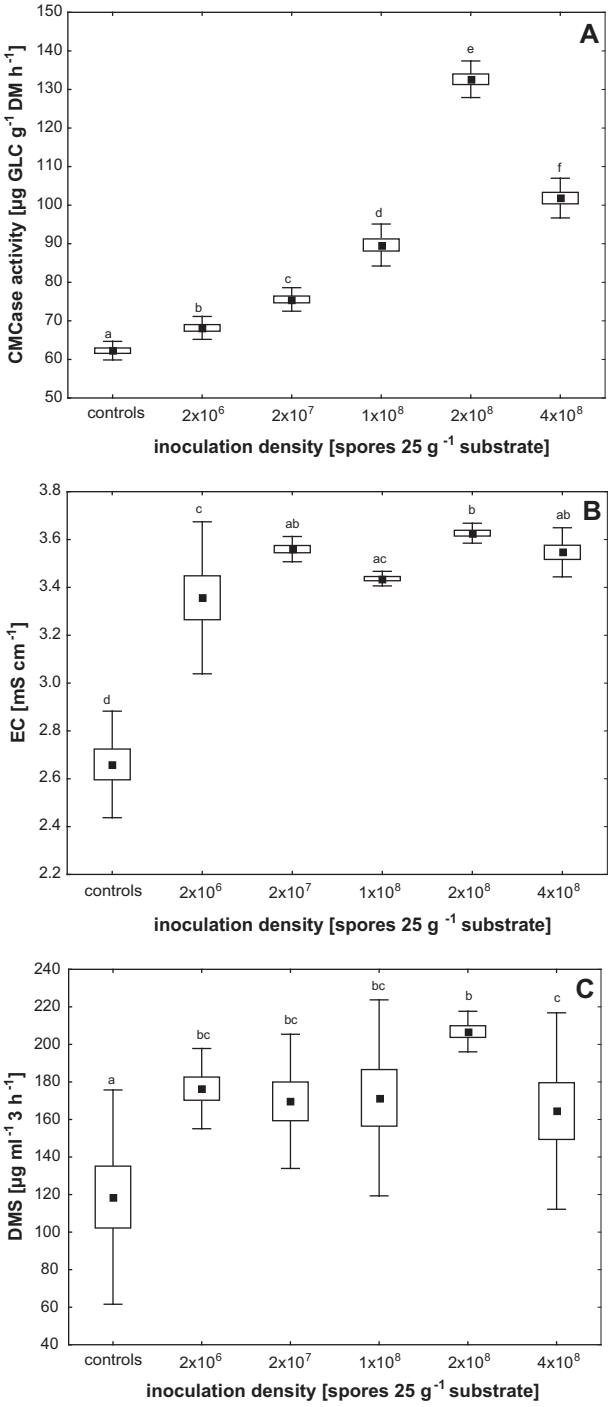



**Fig. 2.** Impact of varying spore densities on (A) CMCase activity after ten days of aerobic pre-treatment, (B) electrical conductivity (EC) and (C) DMSO reductase activity after 14 days of anaerobic incubation. Results are given as means (±1.96 SD), $n = 3$. Significant differences ($p < 0.05$) are indicated by different characters.

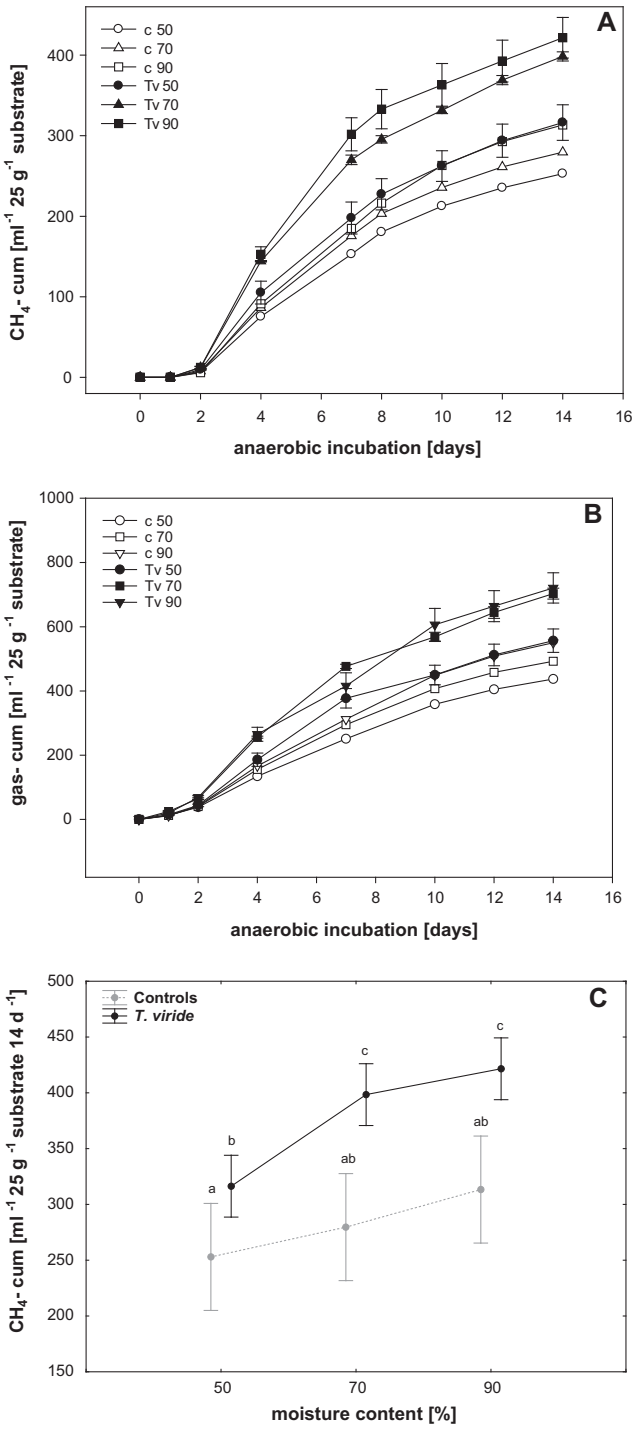



**Fig. 3.** (A) Cumulative methane and (B) gas production rates within 14 days of anaerobic incubation depending upon inoculation with *T. viride* and different initial moisture contents. Data points are means (±SD), $n = 3$. *c 50*, *c 70*, *c 90*; controls with 50%, 70%, and 90% moisture content. *Tv 50*, *Tv 70*, *Tv 90*; *T. viride* with 50%, 70%, and 90% moisture content. (C) Interaction plot of a multivariate ANOVA showing the cumulative methane yields achieved in the inoculated and non-inoculated samples with different moisture contents after 14 days of anaerobic incubation. Significant differences ($p < 0.05$) are indicated by different characters. Bars indicate 95% confidence intervals.

contents (Flannigan et al., 2002). In this context, previous studies have shown that high moisture levels enhanced fungal growth and cellulase production with a prevalence of lignocellulosic substrates serving as carbon sources (Kalogeris et al., 2003), which is in agreement with the results obtained in the present study. Low CMCase activities in relation to low moisture were found to be associated with reduced substrate solubility and swelling capacity causing high water tension (Ghanem et al., 2000). The less distinct increase in the yield of methane in the samples with moisture contents of 50% compared to those with moisture contents of 70% and 90% could therefore be related to insufficient quantities of water affecting nutrient availability, as indicated by the reduced EC, which in turn inversely influenced cell metabolism causing methane rates to cease. While studying the impact of different

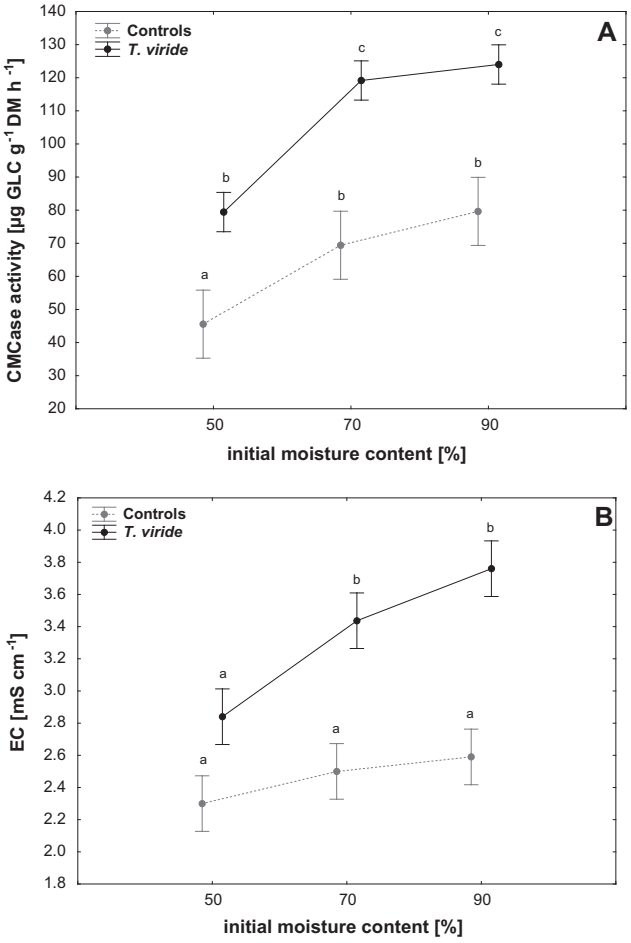



**Fig. 4.** Interaction plots of ANOVA showing the impact of inoculation with *T. viride* and different initial moisture contents on (A) CMCase activity after ten days of aerobic pre-treatment and (B) EC after 14 days of anaerobic treatment. Significant differences ($p < 0.05$) are indicated by different characters. Bars indicate 95% confidence intervals.

water levels on the anaerobic degradation performance of organic matter, Bouallagui et al. (2003) reported that an initial moisture content of 60–80% provided maximum methane production rates, which is consistent with the results obtained in the present study. Due to the highest yield improvement achieved within anaerobic treatment and no significant difference to the samples with an initial water content of 90%, 70% was considered as the optimal level of moisture during aerobic pre-treatment with *T. viride* and subsequent anaerobic digestion.

### 3.3. Impact of the duration of aerobic pre-incubation

The duration of the aerobic pre-incubation turned out to be one of the most critical factors affecting the efficiency of the anaerobic conversion performance, with methane and gas yields significantly increasing along with a progressive pre-incubation time of up to ten days (Fig. 5A and B). Subsequent prolongation of aerobic pre-treatment to twenty days led to an increase in methane yield in the controls, while it declined in the samples supplemented with *T. viride* (Fig. 5A and C). Regardless of the inoculation with *T. viride*, a three-day aerobic pre-treatment led to very low methane yields achieved within anaerobic treatment (Fig. 5A). The underlying causes were attributed to the acid pH-values observed during the first days of aerobic pre-treatment, which in turn led to unfavorable initial conditions for the subsequent anaerobic digestion.

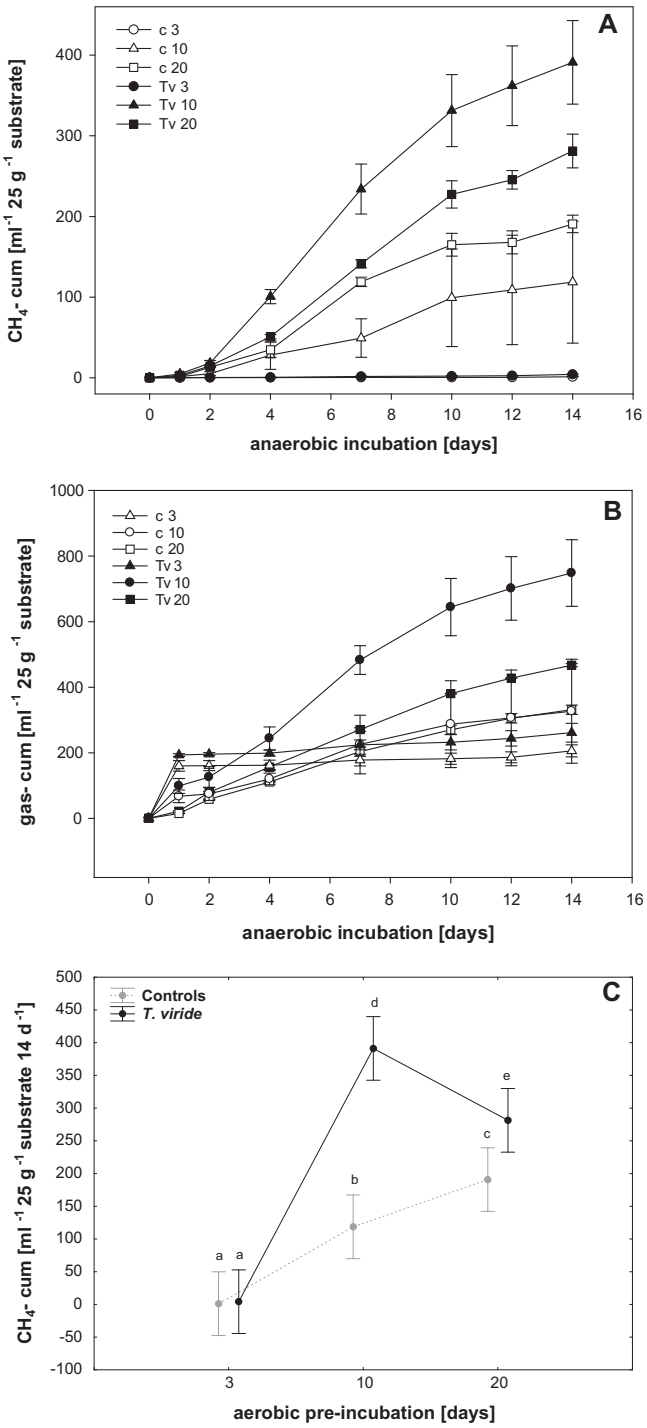



**Fig. 5.** (A) Cumulative methane and (B) gas production within 14 days of anaerobic incubation depending upon inoculation with *T. viride* and aerobic pre-incubation time. *c 3*, *c 10*, *c 20*; controls with a three-, ten, and twenty-day pre-treatment. *Tv 3*, *Tv 10*, *Tv 20*; *T. viride* with a three-, ten, and twenty-day pre-treatment. Data points are means (±SD), $n = 3$. (C) Interaction plot of a multivariate ANOVA showing the impact of pre-treatment with *T. viride* and varied pre-incubation periods on methane yield after 14 days of anaerobic treatment. Significant differences ($p < 0.05$) are indicated by different characters. Bars indicate 95% confidence intervals.

After a three-day aerobic pre-treatment, pH values as low as 4.0 were observed as a result of organic acid production, oxidation of reduced sulfur and nitrogen compounds, and $CO_2$ formed during the initial stages of aerobic organic matter decay. The acidic conditions during pre-treatment might have further affected mycelia

growth causing low CMCase activities (Fig. 6A), since the cellulase system of *T. viride* was found to have its optimum at pH 6.5 (Harman and Kubicek, 1998). While studying the effect of initial pH on the biosynthesis of cellulases by *T. viride*, Malik et al. (2010) also reported very little CMCase production at pH 4.5, which is consistent with our findings. Reduced enzyme activity could be explained by the fact that low pH causes denaturation of the enzymes and may have a regulatory effect on the adsorption and desorption characteristics of cellulolytic enzymes. After a fortnightly anaerobic treatment of the three-day aerobic pre-treated samples pH was still low (pH < 6.2) and outside the range specified for efficient anaerobic degradation performance (Table 3), which resulted in an entirely inhibition of $CH_4$ production (Fig. 5A and C), thus indicating disequilibrium in the metabolic chain and disturbed degradation performance. In contrast to any other measurements, with intermediate volatile fatty acids rapidly being consumed, VFA quantification in the samples that have undergone a three-day aerobic pre-incubation revealed a buildup and persistence of acetate and butyrate whereas propionate was not detectable during anaerobic treatment (data not shown). Inanc et al. (1999) reported that the formation of propionate is suppressed at low pH, suggesting acidic conditions as the primary cause. A buildup of acetate can, in turn, reduce the free energy available for the anaerobic oxidation of metabolic intermediates such as butyrate, leading to intermediate VFA accumulation (Hickey and Switzenbaum, 1991). High VFA conditions along with low pH are known to be inhibitory to methanogenic archaea, resulting in a significant decrease in metabolic activity and thus in declining methane yield (Illmer et al., 2009). An aerobic pre-incubation period of only three days should therefore be considered insufficient due to lacking pH-stabilization, leading to unstable conversion performance.

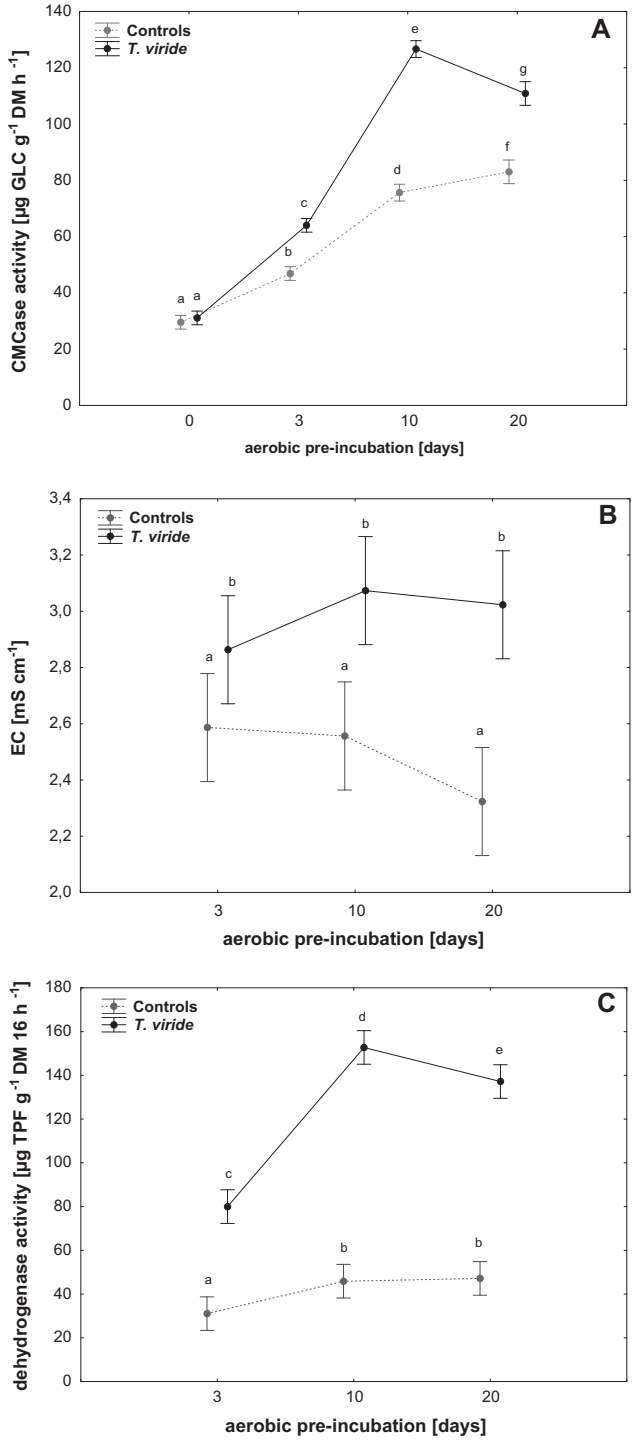



**Fig. 6.** Impact of inoculation with *T. viride* and pre-incubation time on (A) CMCase activity after three, ten and twenty days of aerobic pre-treatment, (B) EC, and (C) dehydrogenase activity after 14 days of anaerobic treatment. Interaction plots of a multivariate ANOVA with 95% confidence intervals. Significant differences ($p < 0.05$) are indicated by different characters.

A ten-day aerobic pre-treatment with *T. viride*, however, led to a steady and highly efficient fermentation process as it was possible to achieve an increase in pH, thus providing suitable conditions for the subsequent anaerobic digestion. Beyond the 3rd day of aerobic pre-treatment, pH started to rapidly increase reaching pH neutral values after ten days of pre-incubation due to the decomposition of organic acids, production of ammonia, and mineralization of organic nitrogen resulting from microbial activity. The increase in pH into the neutral range along with prolongation of pre-incubation favored CMCase production reflected in significantly higher activities compared to the samples with a three-day aerobic pre-treatment (Fig. 6A). CMCase was found to yield maximum activity after ten days of aerobic pre-incubation, with a significant increase in the yield of reducing sugar equivalents ascertained in the samples inoculated with *T. viride* compared to those noted in the corresponding controls (Fig. 6A). Previous time course profiles of CMCase production by *T. viride* identified maximum activity occurring after 7–8 days of incubation (Kandari et al., 2013), which is in agreement with the results obtained in the present study. Neutral pH values and high CMCase activities corresponding to sufficient substrate conversion rates along with facilitated utilization of the carbon compounds contained, as indicated by the significant increase in EC in all samples supplemented with *T. viride* (Fig. 6B), might have caused the highly different methane production rates observed during anaerobic treatment.

After a twenty-day aerobic pre-incubation the positive influence of *T. viride* was still obvious due to significantly higher methane yields achieved in the samples that have previously been inoculated with active fungal spores, however, the increase in the yield of methane and total gas was lower compared to the samples that have undergone a ten-day aerobic pre-treatment with *T. viride* (Fig. 5A and B). The decline in methane yield was assigned to the observed decrease in CMCase activities beyond the 10th day of aerobic pre-treatment, with significantly lower activities ascertained in the samples supplemented with *T. viride* and a twenty-day pre-treatment compared to those with a ten-day pre-treatment (Fig. 6A). The stagnation in the efficacy of enzymatic complexes was found to be inextricably linked to innate structural characteristics of the substrate and/or modifications occurring as saccharification proceeds. Among the major reasons are changes in substrate

accessibility and reactivity, increasing cellulose crystallinity, enzyme inactivation, and a decline in overall enzyme synergism (Bansal et al., 2012). Declining substrate accessibility and reactivity is associated with the exhaustion of easily available cellulose fractions over conversion that lessens nutrient availability, which could be linked to the decrease in EC (Fig. 6B) and metabolic activity (Fig. 6C) in the samples supplemented with *T. viride* and a twenty-day aerobic pre-treatment. Other related factors, such as the accumulation of byproducts and/or certain toxic metabolites released are assumed to have an inhibitory effect on the secretion pathways resulting in the inactivation of secretary machinery of the enzymes (Nochure et al., 1993). Hence, an aerobic pre-incubation period of ten days was deemed to be optimal.

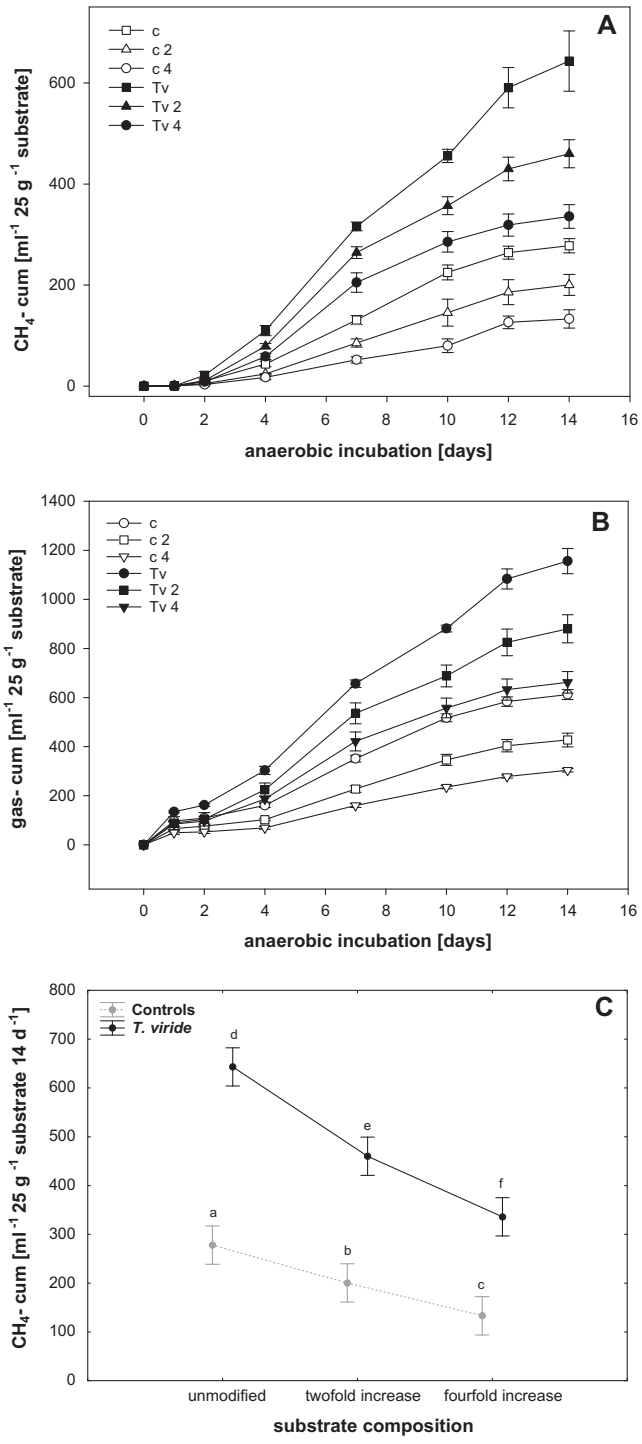



**Fig. 7.** (A) Cumulative methane and (B) gas production within 14 days of anaerobic incubation depending upon inoculation with *T. viride* and the composition of the initial substrate. Data points are means (±SD), $n = 3$. *c*, controls substrate unmodified; *c 2* and *c 4*, controls two- and fourfold increase in wood content, respectively. *Tv, T. viride* substrate unmodified; *Tv 2* and *Tv 4*, *T. viride* two and fourfold increase in the content of wood, respectively. (C) Interaction plot of a multivariate ANOVA showing the impact of pre-treatment with *T. viride* and initial substrate composition on methane yield after 14 days of anaerobic treatment. Significant differences ($p < 0.05$) are indicated by different characters. Bars indicate 95% confidence intervals.

### 3.4. Impact of altering the initial substrate composition

Studies focusing on the impact of the initial substrate composition revealed a positive effect of pre-treatment with *T. viride* on the subsequent anaerobic digestion, with higher methane yields obtained in the samples supplemented with active fungal spores compared to the controls, but also a negative one regarding the modification in substrate composition (Fig. 7A). While the application of *T. viride* resulted in a twofold increase in the yield of methane compared to the controls in the samples in which the composition of the substrate remained unmodified, altering the content of wood was accompanied by a significant diminution in the yield of methane in both, the controls and the samples that were inoculated with *T. viride* (Fig. 7B). The observed decrease in methane production was more distinct the higher the content of

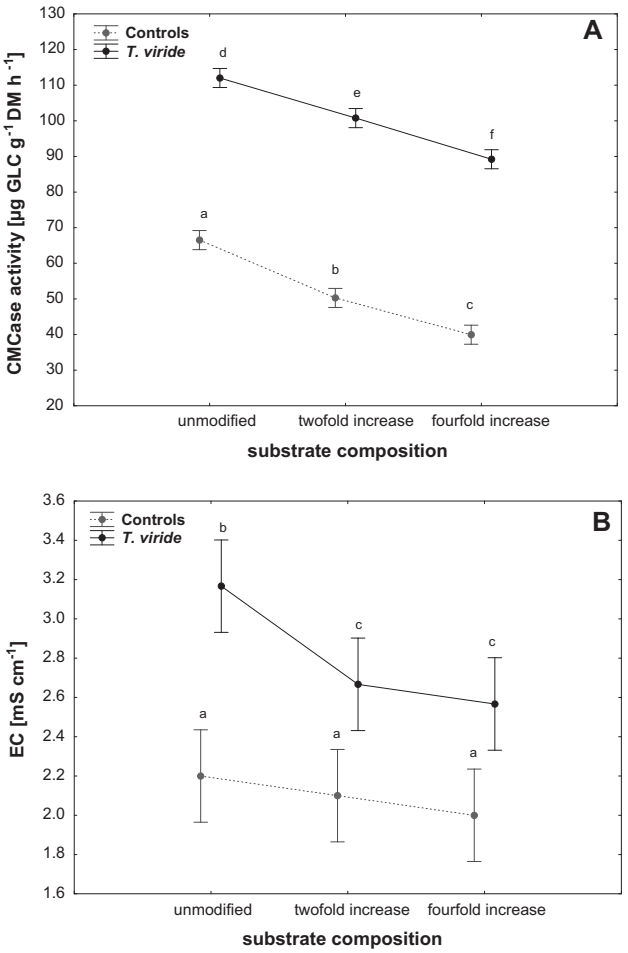



**Fig. 8.** Interaction plots of ANOVA showing the impact of inoculation with *T. viride* and initial substrate composition on (A) CMCase activity after ten days of aerobic pre-treatment and (B) EC after 14 days of anaerobic treatment. Significant differences ($p < 0.05$) are indicated by different characters. Bars indicate 95% confidence intervals.

wood (Fig. 7B) and was most probably related to the persistent as well as recalcitrant character of ligneous substrates with repercussions on the milieu conditions prevailing during the process. A principal reason is that wood based materials contain very low contents of nitrogen, which is required to produce the enzymes needed for sufficient degradation of the main structural components of wood – cellulose, hemicellulose, and lignin. While the samples without any substrate modification exhibited initial C/N ratios of 22–25:1, increasing the content of wood led to higher initial C/N ratios of 35:1. In this context, Eiland et al. (2001) reported that low initial C/N ratios caused more rapid degradation rates of cellulose and hemicellulose (hemicellulose: 50–80%, cellulose: 40–60%), while high initial C/N ratios resulted in 10–20% degradation of both hemicellulose and cellulose. Nitrogen deficiency as reflected by the high initial C/N ratios in the samples with increased wood contents may therefore well have contributed to the gradually decreasing CM-cellulase activities compared to those where the composition of the substrate remained unmodified (Fig. 8A). Low nitrogen substrates may also lack sufficient protein to supply the ammonia needed to neutralize acidic conditions (Haug, 1993), which could be connected to the slight decrease in pH observed in the samples with modified substrate compositions (Table 3). In addition, ligneous substrates exhibit a natural acidity of a slight-to-moderate degree, which might have also caused a drop in pH. Increasing the content of wood further resulted in a gradual decrease in EC (Fig. 8B) in both, inoculated and non-inoculated samples, which could point to an inhibitory effect of higher wood contents due to a limited excess of nutrients. In addition to these points, wood contains natural accompanying substances such as tannins, well known for their ability to cross-link proteins, as well as furan derivatives and phenolic compounds formed and/or liberated during wood decay. It is therefore equally conceivable that these substances exert a potentially inhibitory effect on the interacting microbiocenosis during anaerobic treatment and should therefore be examined further.

## 4. Conclusion and future prospects

Biological pre-treatment of organic matter using *T. viride* in an aerobic upstream process prior to anaerobic digestion led to a significant increase in substrate specific methane and gas yield, with single-parameter variation studies revealing optimal conditions with an inoculation density of $2 \times 10^8$ spores 25 $g^{-1}$ substrate, an initial moisture content of 70%, and an aerobic pre-incubation extending over ten days in total. The improvement in overall efficiency was highly reproducible within test replications, and thus a promising technique to pre-treat (ligno) cellulosic feedstock for enhanced biogas production.

The procedure adopted harbors significant future potential for improving the profitability of biogas plants and represents an economically feasible process to treat domestic organic waste as a valuable material conducive to environmentally friendly recycling economy. The facile digestibility and conversion of lignocellulosic materials based on enzyme initiated cascade reactions involving (ligno) cellulolytic microorganisms therefore provides a significant opportunity to improve the overall efficiency of energetic biomass utilization under anaerobic conditions. On the basis of the insight gained in the present study, further investigations shall therefore ultimately contribute towards identifying the major biochemical and microbiological factors responsible for the sustainable productivity enhancement. Common molecular-biological techniques are intended to provide an insight into microbial diversity and the functional relationships of the dominant phylogenetic groups, thus improving the understanding of metabolic fluxes within anaerobic treatment. In addition, large-scale assessment shall provide the basis for broad practical applications.

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